



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS FALL 2022

Faculty:	Computing & Applied Sciences	Department:	Computer Science & Information Technology	Batch:	2021F
Semester:	3 rd	Degree Program:	BS (Computer Science)	Course Instructor(s):	Mr. Shakir Karim / Dr. Abdul Hameed Pitafi
Course Code:	CS-215T	Course Title:	Computer Organization & Assembly Language	Credit Hours:	3
Date of Examination:	01-02-2023	Timings:	9:30 am to 12:30 pm	Shift:	Morning

Timings: 3 Hours

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll# and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Please return the Question Paper along with the Answer books.

Roll #		Name		Section		'A' Copy Serial No.	
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Q.No.1**[6+4]**

- Differentiate between RISC and CISC processors on the basis of execution time. Also highlight the main characteristics of a RISC processor.
- Consider the problem of swapping the bit pattern in a byte or a word as an application of the shift and rotate instruction. Write the Intel x86 Assembly Language based code that swaps the higher 8bits of a word to the lower 8bits. Assume that the word is stored in BX register and BX contains the value 64BAH.

Q.No.2**[5+5]**

- Multiply $+13_{10}$ by -11_{10} using Booth's algorithm. Show all the necessary steps involved.
- Represent 2419_{10} in 32-bit IEEE 754 floating-point format. Show all the necessary steps involved.

Q.No.3**[5+5]**

Consider a memory system that uses 20-bits to address at the byte level, with a cache memory that uses a line size of 16-byte.

- Assume a direct mapped cache with a line field consists of 8 bits. Show the address format and determine the parameters: *Number of addressable units, Number of bytes in cache memory, Number of lines in cache, Tag size.*
- Assume an associative cache. Show the address format and determine the parameters: *Number of blocks in main memory, Number of bytes in main memory, Block size, Tag size.*



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Q.No.4

[10]

Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and average cycles per instruction of 4. The same processor is upgraded to a pipelined processor with five stages but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume there are no bubbles in the pipeline. Find the speed up achieved in this pipelined processor.

Q.No.5

[10]

Refer to the table I and instruction set summary (for *MOV*, and *ADD* commands), write machine code for the given instructions.

- (i) *ADD AL, [0200]* (iv) *ADD BL, CH*
 (ii) *MOV CL, [BX]* (v) *MOV SS, 1234H*
 (iii) *ADD DX, [BP]*

Table I: R/M and MOD values

R/M \ MOD	00	01	10	11	
	0-bit displacement	8-bit displacement	16-bit displacement	w=0	w=1
000	[BX+SI]	[BX+SI]+D8	[BX+SI]+D16	AL	AX
001	[BX+DI]	[BX+DI]+D8	[BX+DI]+D16	CL	CX
010	[BP+SI]	[BP+SI]+D8	[BP+SI]+D16	DL	DX
011	[BP+DI]	[BP+DI]+D8	[BP+DI]+D16	BL	BX
100	[SI]	[SI]+D8	[SI]+D16	AH	SP
101	[DI]	[DI]+D8	[DI]+D16	CH	BP
110	DIRECT	[BP]+D8	[BP]+D16	DH	SI
111	[BX]	[BX]+D8	[BX]+D16	BH	DI

8088

intel.

8088/8088 Instruction Set Summary

Mnemonic and Description	Instruction Code			
DATA TRANSFER				
MOV – Move:	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0
Register/Memory to/from Register	1 0 0 0 1 0 d w	mod reg r/m		
Immediate to Register/Memory	1 1 0 0 0 1 1 w	mod 0 0 0 r/m	data	data if w = 1
Immediate to Register	1 0 1 1 w reg	data	data if w = 1	
Memory to Accumulator	1 0 1 0 0 0 w	addr-low	addr-high	
Accumulator to Memory	1 0 1 0 0 0 1 w	addr-low	addr-high	
Register/Memory to Segment Register	1 0 0 0 1 1 1 0	mod 0 reg r/m		
Segment Register to Register/Memory	1 0 0 0 1 1 0 0	mod 0 reg r/m		
ARITHMETIC				
ADD – Add:	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0
Reg./Memory with Register to Either	0 0 0 0 0 d w	mod reg r/m		
Immediate to Register/Memory	1 0 0 0 0 s w	mod 0 0 0 r/m	data	data if s w = 01
Immediate to Accumulator	0 0 0 0 0 1 w	data	data if w = 1	